

allowing the body's cells to absorb and use glucose. Without it, glucose cannot enter the cell and therefore cannot be used as fuel for the body's functions.^[90] If the pancreas is exposed to persistently high elevations of blood glucose levels, the insulin-producing cells in the pancreas could be damaged, causing a lack of insulin in the body. Insulin resistance occurs when the pancreas tries to produce more and more insulin in response to persistently elevated blood glucose levels. Eventually, the rest of the body becomes resistant to the insulin that the pancreas is producing, thereby requiring more insulin to achieve the same blood glucose-lowering effect, and forcing the pancreas to produce even more insulin to compete with the resistance. This negative spiral contributes to pancreatic burnout, and the disease progression of diabetes.

To monitor the body's response to blood glucose-lowering therapy, glucose levels can be measured. Blood glucose monitoring can be performed by multiple methods, such as the fasting glucose test which measures the level of glucose in the blood after 8 hours of fasting. Another test is the 2-hour glucose tolerance test (GTT) – for this test, the person has a fasting glucose test done, then drinks a 75-gram glucose drink and is retested. This test measures the ability of the person's body to process glucose. Over time the blood glucose levels should decrease as insulin allows it to be taken up by cells and exit the blood stream.

Hypoglycemia management

Individuals with diabetes or other conditions that result in low blood sugar often carry small amounts of sugar in various forms. One sugar commonly used is glucose, often in the form of glucose tablets (glucose pressed into a tablet shape sometimes with one or more other ingredients as a binder), hard candy, or sugar packet.



Glucose, 5% solution for infusions

Sources

Most dietary carbohydrates contain glucose, either as their only building block (as in the polysaccharides starch and glycogen), or together with another monosaccharide (as in the hetero-polysaccharides sucrose and lactose).^[91] Unbounded glucose is one of the main ingredients of honey.



Glucose tablets

Table 1. Sugar content of selected common plant foods (g/100g)^[92]

Food Item	Total Carbohydrate ^A including "dietary fiber"	Total Sugars	Free Fructose	Free Glucose	Sucrose	Fructose/ Glucose Ratio	Sucrose as a % of Total Sugars
Fruits							
Apple	13.8	10.4	5.9	2.4	2.1	2.0	19.9
Apricot	11.1	9.2	0.9	2.4	5.9	0.7	63.5
Banana	22.8	12.2	4.9	5.0	2.4	1.0	20.0
Fig, dried	63.9	47.9	22.9	24.8	0.9	0.93	0.15
Grapes	18.1	15.5	8.1	7.2	0.2	1.1	1
Navel orange	12.5	8.5	2.25	2.0	4.3	1.1	50.4
Peach	9.5	8.4	1.5	2.0	4.8	0.9	56.7
Pear	15.5	9.8	6.2	2.8	0.8	2.1	8.0
Pineapple	13.1	9.9	2.1	1.7	6.0	1.1	60.8
Plum	11.4	9.9	3.1	5.1	1.6	0.66	16.2
Vegetables							
Beet, Red	9.6	6.8	0.1	0.1	6.5	1.0	96.2
Carrot	9.6	4.7	0.6	0.6	3.6	1.0	77
Red Pepper, Sweet	6.0	4.2	2.3	1.9	0.0	1.2	0.0
Onion, Sweet	7.6	5.0	2.0	2.3	0.7	0.9	14.3
Sweet Potato	20.1	4.2	0.7	1.0	2.5	0.9	60.3
Yam	27.9	0.5	traces	traces	traces	na	traces
Sugar Cane		13–18	0.2 – 1.0	0.2 – 1.0	11–16	1.0	high
Sugar Beet		17–18	0.1 – 0.5	0.1 – 0.5	16–17	1.0	high
Grains							
Corn, Sweet	19.0	6.2	1.9	3.4	0.9	0.61	15.0

^{^A} The carbohydrate figure is calculated in the USDA database and does not always correspond to the sum of the sugars, the starch, and the "dietary fiber".